university
of south africa
PRESS

Article title: Educational Data Mining in Open Distance e-Learning: A Pragmatic Approach to Module-level Analytics
Authors: Kelly Anne Wilken[1], Jaroslaw Adamiak[2]
Affiliations: university of south africa[1], student retention unit, university of south africa[2]
Orcid ids: 0000-0003-2801-0270[1], 0000-0001-6535-6831[2]
Contact e-mail: youngka@unisa.ac.za
License information: This work has been published open access under Creative Commons Attribution License http://creativecommons.org/licenses/by/4.0/, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. Conditions, terms of use and publishing policy can be found at https://www.scienceopen.com/.
Preprint statement: This article is a preprint and has not been peer-reviewed, under consideration and submitted to UnisaRxiv for open peer review.
DOI: 10.25159/UnisaRxiv/000030.v2
Preprint first posted online: 28 January 2025
Keywords: Educational data mining (EDM), Knowledge discovery in databases (KDD), Open distance e-learning (ODeL), At-risk students, South Africa

Educational Data Mining in Open Distance e-Learning: A Pragmatic Approach to Module-level Analytics

Kelly Anne Young
https://orcid.org/0000-0003-2801-0270
University of South Africa
youngka@unisa.ac.za

Jaroslaw Adamiak
https://orcid.org/0000-0001-6535-6831
University of South Africa
eadamijp@unisa.ac.za

Abstract

In this article, we explore whether educational data mining (EDM) processes, supported by commercial software and used by a non-specialist member of academia, can easily be applied to solving common higher education problems, such as identifying potentially at-risk students in need of intervention or remediation. In particular, the authors advocate for the knowledge discovery in databases (KDD) process (adjusted to the academic context), which comprises nine steps. Using these nine steps, the authors demonstrate a pragmatic approach to module-level learner analytics using a first-year calculus course at a leading open distance e-learning institution in the Global South.

Keywords: educational data mining (EDM); knowledge discovery in databases (KDD); open distance e-learning (ODeL); at-risk students; South Africa

Introduction

Data-intensive science has been recognised as the fourth research paradigm, after the experimental, theoretical and computational ones (Bell, Hey, and Szalay 2009). Data mining (DM), a methodology that adheres to that paradigm, is the process of discovering knowledge or extracting hidden patterns from massive amounts of pre-existing data, leading to novel insights, new applications and/or alternative interpretations of institutional processes (Fynn and Adamiak 2018; Liu and Özsu 2018). The concept of DM arose in response to the substantial growth in computer processing abilities and a huge increase in the volume of gathered data, mostly in an electronic format.

Educational Data Mining (EDM) applies DM procedures to educational data and is concerned with developing methods to improve the understanding of students and the context(s) in which they learn (Bakhshinategh et al. 2018; Romero and Ventura 2010). EDM methods integrate concepts and applications from a wide area of research, including machine learning, psychometrics, statistics, information visualisation, and computational modelling (Salinas and Stephens 2015). Evidently, EDM encompasses a range of applications and tasks that contribute to enhancing the educational experience. According to Romero et al. (2011), these include:

1. **Facilitating stakeholder communication:** In this regard, EDM supports effective communication between course administrators, lecturers, and other stakeholders by providing valuable insights into students' activities and usage information within courses. This information enables administrators and lecturers to gain a deeper understanding of student behaviour and make informed decisions regarding course management.
2. **Enhancing course maintenance and improvement:** By leveraging information about student usage and learning, EDM assists course administrators and lecturers in identifying areas for improvement. The use of this data allows higher education stakeholders to make informed decisions and implement changes that enhance the overall quality of the courses, resulting in a better learning experience for students.
3. **Personalised content recommendations:** By leveraging data on student usage patterns and preferences, EDM can generate personalised content recommendations. By analysing students' activities and identifying their current needs, the system can suggest the most relevant and appropriate content to individual students, enhancing their learning outcomes and engagement.
4. **Predicting student grades and learning outcomes:** Utilising data derived from students' course activities, EDM can predict final grades or other types of learning outcomes. This predictive capability helps educators gain insights into student progress, identifies students at risk of falling behind, and takes proactive measures to support their learning journey.
5. **Student modelling for course development and support:** EDM plays a crucial role in student modelling, which involves creating comprehensive profiles of students. By analysing various aspects of student behaviour, learning styles, and performance, EDM enables course developers and educators to design more tailored and effective learning experiences for particular profiles of students. This modelling process guides course development and supports initiatives that cater to the diverse needs of students.

Overall, EDM can personalise the learning experience in distance education (Y. Xu, Zhang, and Gao 2020); provide recommendations for students; predict student performance in open distance learning (Alachiotis et al. 2022; Aydoğdu 2020); provide models of student profiles; detect undesirable student behaviours or learners at-risk (Bendjebar, Lafifi, and Seridi-Bouchelaghem 2022); group students by predefined criteria; or develop concept maps of learning programmes to inform design (Salinas and Stephens 2015). An additional benefit of EDM lies in answering queries on forecasting, object classification, associating similar events, clustering objects into groups, and discovering causal relationships within educational contexts (Pal 2011). The study that informed this article explored whether EDM processes, supported by commercial software and used by a non-specialist member of academia, can easily be applied to solving common higher education problems, such as identifying potentially at-risk students in need of intervention or remediation. In this context, an at-risk student is one whose academic and demographic data exhibit patterns similar to historical data (i.e., to

those students who have failed the course in previous years). Our research questions were:

- **Q1**: Is there an EDM framework that can (easily) be transferred to ODeL settings to identify potentially at-risk students?
- **Q2:** Can this framework be used by a lecturer, researcher, or non-specialist member of academia who is not necessarily a data science specialist?

Tools and Processes

Although computing skills are required for data-intensive research, expert-level abilities are generally not necessary, in part because of the availability of pre-existing software packages and libraries that implement advanced techniques (Chen and Wojcik 2016). Within academia, the programming language "Python" (https://www.python.org/) gained substantial popularity, together with "Pandas" (https://pandas.pydata.org/)—a data manipulation library, and "Scikit-learn" (https://scikit-learn.org/)—a package containing a range of machine learning algorithms. Within psychology, on the other hand, statistical programming alternatives, such as "R" (https://www.r-project.org/), together with a library of precompiled packages (https://cran.r-project.org/), are often preferred.

Amongst the dedicated DM software applications, "Waikato Environment for Knowledge Analysis" (WEKA) and the "Statistical Package for the Social Sciences" (SPSS) are often adopted (see Slater et al. 2017 for a comprehensive list of DM applications and environments). For the purposes of the current study, however, the authors utilised RapidMiner (https://rapidminer.com/platform/educational/)—a data science platform designed for enterprises that analyse the collective impact of their employees, expertise and data. RapidMiner Studio, according to the company owners, is available for academics looking for an end-to-end data science platform for instructional or research purposes.

There are a number of different frameworks for DM, the most popular of which include Knowledge Discovery in Databases (KDD); Cross-Industry Standard Process for Data Mining (CRISP-DM); Sample, Explore, Modify, Model, Assess (SEMMA) (Shafique and Qaiser 2014) and the Obtain, Scrub, Explore, Model, and iNterpret (OSEMN) framework (Mason and Wiggins 2010). While they share similarities in terms of their overall process, the differences between these frameworks are not substantial (Azevedo and Santos 2008), differing only in terms of their specific stages and focus.

Discovery Framework Application

We adopted the KDD framework as it is most closely aligned with the scientific inquiry process. Moreover, it is widely used in academia (Kurgan and Musilek 2006). The KDD process, adjusted to the academic research environment, is outlined below in nine steps (Maimon and Rokach 2011):

1. Developing an understanding of the research problem and inherent research questions.
2. Selecting data and creating a data set on which the mining will be performed.
3. Data pre-processing.
4. Data transformation.
5. Choosing the appropriate (E)DM task.
6. Choosing the (E)DM algorithm.
7. Employing the (E)DM algorithm.
8. Evaluation.
9. Using the discovered knowledge.

These nine steps are outlined below, with an illustration of each.

Step 1: Developing an understanding of the research problem and inherent research questions

During the first step, members of academia (i.e., lecturers, researchers, administrators, etc.) need to decide what task the EDM is intended to accomplish. The goals must be defined in terms of the research questions and then reformulated as a DM problem statement. Typically, this problem statement will contain keywords such as prediction, classification, association, estimation, grouping, clustering, and so forth. As DM is performed in a chosen application domain, this domain-specific knowledge is important in order to devise a meaningful problem statement (Cherkassky and Mulier 2006).

One of the frequently asked questions posed by module leaders is the quantification of student success in their respective courses. Many lecturers state that knowing the outcome of a course examination (particularly within high- or at-risk modules) allows for the early identification of at-risk students, the contributing factors, and the potential provision of remediation and interventions. By accurately predicting student performance in a timely manner, lecturers can identify underperforming students and take proactive measures to intervene and implement the necessary steps during the learning process (Namoun and Alshanqiti 2021).

Within the confines of the University of South Africa (UNISA), a number of modules were identified as “at-risk” based on certain criteria (such as registration figures, attrition rates, and course success rates) (Barnes 2014), with some modules demonstrating dismal course success rates. One such module is a first-level calculus module (i.e., MAT1512), with course success rates consistently hovering between 34.5% in 2013 to 49.4% in 2021 (Uni-telligence Data Hub 2022). Clearly, nowhere is the identification of at-risk students more topical and pressing than in this particular module, which has demonstrated weak throughput rates for almost a decade and, being on the critical patch for attaining a bachelor’s degree, cannot be skipped by the students. Resultantly, our DM task was to develop a model that could predict the outcome of future course examinations (MAT1512) using historical data.

Step 2: Selecting data and creating a data set on which the mining will be performed

During the second step, members of academia must collect, organise, and inspect the data pertinent to the research problem. This may involve querying institutional databases, web harvesting, downloading relevant files, scanning documents, or importing electronic logs from digital systems (Aggarwal 2015). It is important to note that the data may be structured or unstructured and provided in a variety of formats. An important element here is the recognition of data ownership and the ethical issues surrounding the protection of personal information, as contained in the POPI Act (https://popia.co.za/).

When implementing this step, the researchers contacted the relevant research ethics and research permission departments and completed the required applications for acquiring permission to use the module data. These departments, after detailed scrutiny of the applications, granted us permission to use historical data gathered from MAT1512 students, specifically those registered for the module from 2010–2018 (Reference #: 2017_RPSC_071_AR). This period choice was dictated by a relative temporal proximity to the present time, and not being affected by potential Covid-generated anomalies. Once granted, specific data were requested from the Information and Communications Technology (ICT) department at the institution. The requested meta-data were relevant to the problem under study and contained the variables tabulated below. The data were stored in a CSV[1] format file (in MS Excel) and given the file name "MAT1512Results.csv".

Table 1: Variables used to develop a model of at-risk students in MAT1512

Assignment 1 mark (%)	Home language
MAT1512 Examination outcome	Matric aggregate
Age	Financial aid status
Disability status	Years registered at UNISA

The resulting dataset comprised 2,473 students, along with their respective demographic characteristics, the mark obtained on the first assignment (expressed as a percentage), and the outcome on the MAT1512 examination. As far as the latter is concerned, the following categorical outcomes were possible; passed, failed, passed with distinction, absent, supplement, and cancelled. In RapidMiner, this outcome variable is known as the "label" variable (https://rapidminer.com/).

Step 3: Data pre-processing

Usually, the data obtained in the previous step are not ready for processing but may be incomplete, duplicated, corrupt, irrelevant, badly formatted or inaccurate

1 Comma-separated values.

(www.tableau.com/learn/articles/what-is-data-cleaning). In order to fix these issues, data-cleaning processes are applied, and these typically involve the detection of anomalies, removal of rows with missing critical data, and inputting of missing data where possible (García, Luengo, and Herrera 2015). Details of the data-cleaning procedure depend on the quality and type of data, and the adopted EDM software that contains appropriate procedures. The most common data problems include unusual values (i.e., outliers), missing values, and exact or near duplicates.

Students who did not submit their first assignment had no data entry in the assignment column; this was displayed as "null" in the dataset. Since this variable's missing values would not add any insight into the model's development, the data pertaining to these students were removed from the dataset (i.e., listwise deletion). Although some students had no entry in the outcome column (i.e., no examination result), this data were used to apply the model and determine its predictive power (n = 449).

Next, we ran a statistical and graphical analysis with the aim of identifying outliers, anomalies and any other missing values. These results revealed that a large portion of the students had missing matric aggregates (n = 705). Although decision trees are reportedly "robust even in the presence of missing values" (Song and Lu 2015, 130), these missing values were subsequently filtered out when employing the algorithm. Further scrutiny of the analysis revealed additional missing values in two more variables (i.e., financial aid status and the number of years registered at UNISA); however, the missing data were minimal in both variables (n = 1) and thus retained in constructing the model. After the data-cleaning procedure mentioned here, the resulting sample contained the data of 1,437 MAT1512 students.

Step 4: Data transformation

The clean data derived in step three may still be improved by transforming it into a more digestible format (Aggarwal 2015). Typically, big datasets are split into easily manageable chunks and normalised in order to remove magnitude bias.[2] During this step, and considering that many algorithms require numerical rather than text inputs, one would traditionally transform any categorical variables into numerical ones (this is known as encoding). In addition, when a dataset has too many variables, dimension reduction techniques are useful in minimising the dataset to a limited number of composite scores using feature selection and extraction processes (see Field 2018 for dimension reduction techniques in SPSS). Another way to safely limit big datasets is aggregation, where similar rows of data are combined. When aggregation is impossible, a representative sample is selected rather than all the data.

2 Data normalisation processes involve scaling the numerical attribute values to align the intervals. In doing so, same or equal importance can be assumed (Zhao and Cen 2013). Failure to normalise may lead to attributes (containing large numbers) being treated preferentially by some MLs—this is also known as the magnitude bias.

The data in our dataset appeared to be clean and simple enough at this point. However, when discussing the model, it was decided that the outcome column (i.e., the examination result) be collapsed to represent only two categories: “Passed” when the student passed the exam or passed it with distinction, or “Failed” in all the other cases. This dichotomous variable was created, which was named “Result” in the CSV file and populated with the help of MS Excel’s “if then” conditional statements. At this point, our data were ready for mining tasks.

The top 10 lines of our dataset are tabulated in table 2 to provide an indication of the variables used in the data mining task. Numerical data included the mark on the first assignment (percentage), a student’s age, their matric aggregate, and the number of years they had been registered at the institution. Categorical data comprised a student’s disability status, their home language, their financial aid status and the dichotomous result on the final MAT1512 examination.

Table 2: The top ten lines of the MAT1512 dataset

Assignment 1 (%)	Age	Disability	Home language	Matric aggregate	Financial aid status	Years at Unisa	Result
85	65	N/A	Afrikaans/English	848	No	2	Passed
91	27	N/A	English	1599	No	2	Passed
60	35	N/A	English	1160	No	1	Failed
35	34	N/A	Sesotho	730	No	12	Passed
73	34	N/A	Afrikaans/English	1640	No	2	Passed
85	36	N/A	Setswana	1480	No	2	Passed
38	53	N/A	IsiZulu	650	No	9	Passed
41	36	N/A	English	1400	No	2	Failed
38	30	N/A	Northern Sotho	1128	No	1	Failed
51	32	N/A	IsiZulu	828	Yes	2	Passed

Step 5: Choosing the appropriate EDM task

The two broad groups of DM tasks are defined as predictive and descriptive (Jain and Srivastava 2013). Predictive tasks allow one to find the location/group to which an object belongs (classification) or the value of an unknown label (regression). On the other hand, descriptive tasks discover relations between attributes (association) or create groupings that can be imposed on objects (clustering).

The main purpose of our project was to develop a data mining model that could be used to identify potentially at-risk students, so our task was predominately predictive in nature. However, if we were to discover some relationships between the attributes that may shed light on the uncovered mechanisms ruling higher education processes, we could use this opportunity to create a hypothesis that could be tested and potentially generalised at a later stage.

Step 6: Choosing the EDM algorithm

The numerical procedures utilised in EDM are called Machine Learning (ML) algorithms, as they emulate the learning process to some extent (Janiesch, Zschech, and Heinrich 2021). An algorithm can belong to one of two groups: supervised or unsupervised. Supervised ML algorithms require the definition of the target variable (or "label variable" in RapidMiner), whereas unsupervised algorithms look for hidden patterns in the data without a particular preference for the relationships amongst the variables in the dataset (Aggarwal 2015).

Unsupervised ML algorithms consist of two major groups: clustering and association. Clustering rests on identifying groups of similar objects in the dataset, whereas the aim of association is to discover relationships between the objects in the dataset. As mentioned, there is no class variable in unsupervised ML approaches, and as a result, unsupervised learning is a tool that is often used to elicit new patterns or relationships in the data (Mccue 2015).

Supervised ML algorithms, on the other hand, are used to uncover hidden relationships between a collection of values obtained from historical data. These data are often organised into a set of predictor variables and a class variable (whose values we attempt to predict—the "label" variable in RapidMiner). While supervised ML algorithms do not appear to differ from common statistical techniques, the statistical model (often described by a mathematical function) is assumed before the adjustment of the models' parameters, whereas supervised ML algorithms construct the model without an assumption regarding the shape of the function. Supervised ML algorithms can be subdivided into classifications, where the output is a categorical variable; and regressions, where the output variable is continuous in nature. The most widely used supervised algorithm in EDM is the decision tree, which builds a model in the form of an inverted tree structure. During this EDM procedure, the dataset is partitioned along the attribute values in such a way as to maximise the information gain measured by

entropy or the Gini index (Witten and Frank 2005). The process is repeated iteratively until the subsets are as homogeneous as possible. The size of the tree can be restricted by pruning (i.e., getting rid of overfitting branches) and/or by stopping rules (those which define the predictability of the model (Attewell and Monaghan 2019).

During the sixth step, the ML algorithm is chosen, as is the software, which will model the data. Considering the predictive nature of our DM task and the pre-existing target or label variable (i.e., passed/failed), a supervised ML algorithm was adopted. Among the various algorithms available, it has been noted that classification trees are considered the most effective method for making predictions when working with demographic data, particularly categorical information like student data (Yasmin 2013).[3]

Classification trees utilise a process of repeatedly dividing the data into more homogeneous groups, allowing for interactive exploration, description, and prediction of patterns and processes. This is achieved by incorporating combinations of explanatory variables to explain the variation of a single response variable (Yasmin 2013). The final result is a tree with decision nodes and leaf nodes. The topmost decision node in a tree corresponds to the best predictor and is called the root node (Lagman and Ambat 2015).

Classification trees offer numerous benefits, which include: (1) the ability to handle various types of data such as numeric, categorical, ratings, and survival data; (2) being unaffected by monotonic transformations of explanatory variables; (3) straightforward and reliable construction process; (4) ease of interpretation; and (5) the capability to handle missing values in both the response and explanatory variables (Yasmin 2013). Given these advantages, the decision tree was adopted as the preferred model. The tree-like structure depicting the outcome of the algorithm is shown in figure 1.

3 If, however, the data differ, a range of algorithms could be tested to determine which produces the best results, with the optimal algorithm ultimately being adopted to model the data.

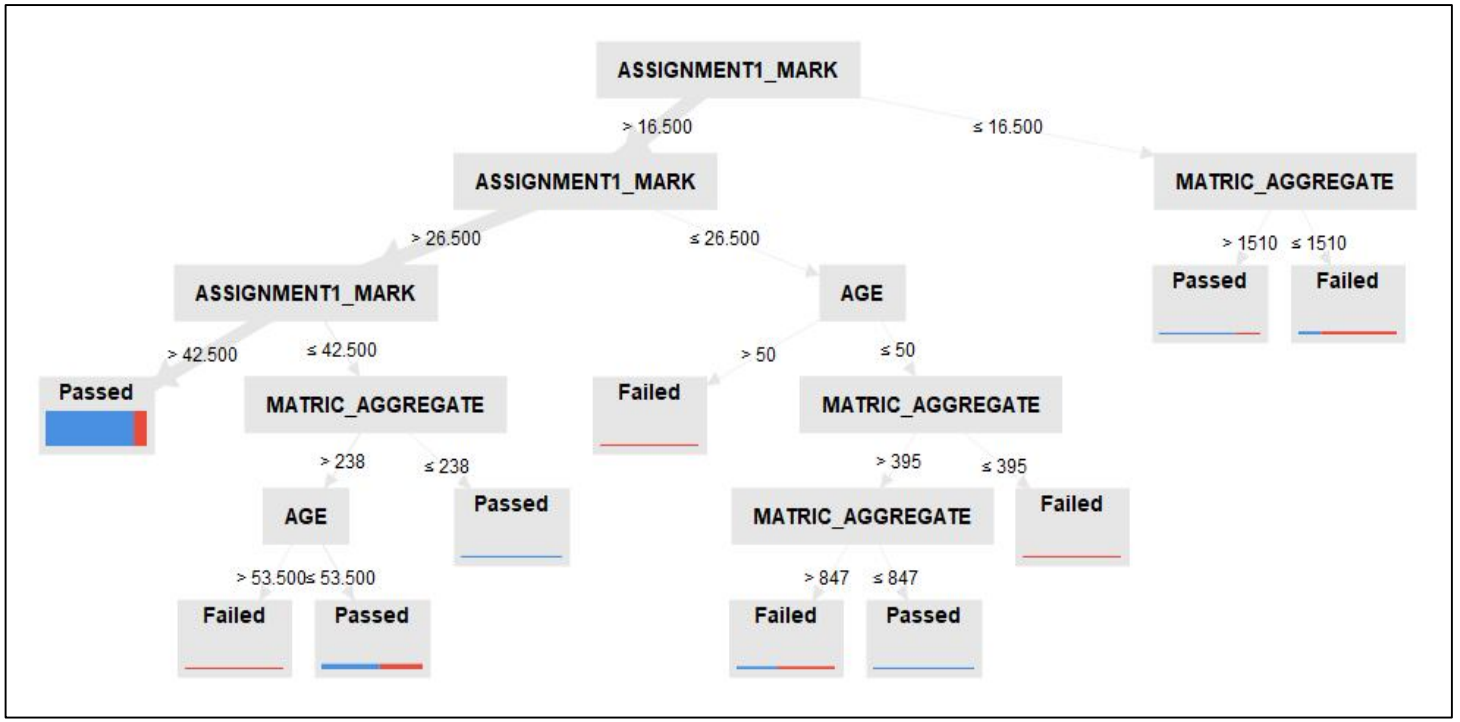


Figure 1: Decision tree generated by RapidMiner using MAT1512 assignment and demographic data

Step 7: Employing the EDM algorithm

During this stage of the KDD process, the classification tree model (which was created in the previous step) needs to be tested or validated. In order to obtain a good fit of the model, we must first split our dataset according to the following definitions (Ripley 2014):

- Training set: A set of examples used to fit the parameters of the model (i.e., the data used to build the model).
- Testing set: A set of examples used to assess the performance of the model (i.e., the data used to validate or test the model built from the training set).

The simplest method of creating training and testing sets can be implemented by randomly assigning 70% of the data to the training set, and the other 30% to the testing set. The numerical value of the split percentage is usually discovered by running a number of experiments.

A more advanced method is k-fold cross-validation (adopted in this study). In k-fold cross-validation, the k denotes the number of equal-sized subsamples we split our dataset into. If we choose k = 10, this means that we will randomly split the dataset into 10 equal-sized subsamples. Of the 10 subsamples, one is retained as the validation data for testing the model, and the remaining subsamples are used as training data. In the second iteration, we promote a different subset to be the testing set and the model is fitted (i.e., tested) again. This process is repeated k times (in our example, 10 times) until all the subsets play a role in the training and testing set. The k results (i.e., the 10 results derived from the cross-fold validations) can then be averaged to produce a single estimation. An inherent advantage of this method is that all observations are used for both training and validating the model.

Step 8: Evaluation

The results produced by the decision tree must be evaluated regarding the suitability towards the solution of the DM problem identified in Step 1 (in this study, i.e., identifying potentially at-risk students). Here, it is particularly important to ascertain which parameters of the algorithm are deemed to be optimal. To achieve this, we need quality measures to apply to the results. One of the most commonly used measures for classification algorithms is the analysis of the confusion table, which gives us the count of correct (i.e., true positives and true negatives) and incorrect behaviours of the algorithm (i.e., false positives and false negatives) when applied to the test dataset.

After completion of the 10-fold cross-validation, results from the confusion matrix revealed that the model demonstrated an overall accuracy of 81.84% (with a standard deviation [SD] of 2.67%). Notwithstanding the relatively high accuracy produced, detailed accuracy by class (shown in table 3) demonstrates a more detailed per/class breakdown of the classifier's prediction accuracy. Of the 283 students who had failed,

the model only correctly classified 62 students (21.91%). Of the 1,154 students who had passed their MAT1512 exam, the model correctly classified 1,114 students (96.53%).

Table 3: Confusion matrix

	True Passed	**True Failed**	**Pred. Totals**	**Class Precision**
Pred. Passed	1114 (True Positives)	221 (False Positives)	1,335	83.45%
Pred. Failed	40 (False Negatives)	62 (True Negatives)	102	60.78%
True Totals	1,154	283		
Class recall	96.53%	21.91%		

Using interpretation guidelines, it is evident that the model does better at predicting a positive outcome than a negative one. To reiterate, among those students who the model placed in the pass category (n = 1,335), the probability of the model being correct stood at 83.45% (this is the positive predictive value, PPV). Lamentably, of those who were placed in the fail category by the model (n = 102), the probability that the model is correct was calculated at 60.78% (this is the negative predictive value, NPV).

Step 9: Using the discovered knowledge

The knowledge gained from the DM process may be used in several ways, including as a law-governing relationship between the data attributes (a part of a new theory) or as an algorithm to be implemented for gaining knowledge from new, incoming data.

Having built our model and tested it (using 10-fold cross-validation), we used the discovered knowledge and applied our model to new, incoming data. For the purposes of illustration, we applied our model to those students who had no data in the examination column and attempted to predict that outcome (n = 449). Table 4 houses a "prediction" column; this column lists the prediction for each student along with a confidence level (for each of the two predictions; pass or fail). These are in the next two columns. We can see, for example, that the model would predict (with 87% confidence) that the first student would pass the examination and thus not be targeted for specific intervention. By contrast, the model predicts that the last student in the table will probably fail the examination, exhibiting a 58% confidence level that the model has correctly classified this student's outcome on the MAT1512 examination. Tailor-made support could then be conceptualised to intervene and decrease the chances of a student failing the module.

Table 4: An example of the student list with prediction field

Prediction	Confidence (Pass)	Confidence (Fail)	Assignment 1 %	Age	Disability	Home language	Matric aggregate	Financial aid status	Years at UNISA
Pass	0.87	0.13	49	29	N/A	Afrikaans	1305	No	1
Pass	0.87	0.13	99	28	N/A	English	1302	No	4
Pass	0.87	0.13	98	58	N/A	IsiXhosa	1300	No	5
Pass	0.87	0.13	100	31	Neurological	English	1293	No	1
Fail	0.42	0.58	19	30	N/A	IsiZulu	1293	No	2
Pass	0.87	0.13	56	32	N/A	IsiZulu	1293	No	8
Pass	0.58	0.42	39	34	N/A	Setswana	1290	No	1
Pass	0.87	0.13	79	33	N/A	Xitsonga	1287	No	1
Pass	0.87	0.13	80	30	N/A	Tshivenda	1281	No	3
Pass	0.87	0.13	97	35	N/A	English	1280	No	4
Pass	0.87	0.13	55	33	N/A	Afrikaans	1280	No	1
Fail	0.42	0.58	23	34	N/A	Afrikaans/Eng	1280	No	1

Discussion

In this study, we presented the use and application of the KDD framework to identify potentially at-risk students in a first-level calculus module at an ODeL institution. We followed the steps outlined in the framework by Maimon and Rokach (2011), which entailed an understanding of the research problem to utilise the discovered knowledge. The framework allowed the researchers to select and pre-process the relevant data, choose an appropriate algorithm, and evaluate the model's performance.

The successful implementation of the KDD framework suggests that it can be easily transferred to an ODeL setting for identifying potentially at-risk students. The framework provides a systematic and structured approach to data mining, which can be adapted and applied to educational data. By following the steps of the framework, educators can effectively analyse educational data, identify patterns, and make predictions related to students' academic performance and risk factors. Thus, we claim that the first research question: *Is there an EDM framework that can (easily) be transferred to ODeL settings to identify potentially at-risk students?* can be answered affirmatively. However, it is important to note that while the KDD framework provides a general structure for data mining, the specific implementation may vary depending on the educational context and the data available. Customisation and adaptation of the framework may be necessary to address the unique challenges and requirements of different educational settings or different countries.

In order to answer the second research question: *Can this framework be used by a lecturer who is not necessarily a data science specialist?* we noted that one of the authors, who was a researcher with a psychology background and not a data scientist, successfully used the KDD framework for this important EDM task. This indicates that individuals with domain-specific expertise, such as educators or lecturers, can successfully utilise the framework without extensive knowledge of data science. A commercial software application, RapidMiner, which is designed to provide an end-to-end data science platform for instructional or research purposes, offers user-friendly interfaces and pre-built algorithms, making it easier for non-specialists to apply data mining techniques without needing to delve into the intricacies of programming or advanced statistical concepts. Thus, we are inclined to answer our second research question also in an affirmative way.

Limitations and Recommendations for Future Directions

No research is without limitation, and this study is no exception. Firstly, the results of the study are exclusively drawn from one module in one institution over a period of years. While undoubtedly useful, particularly to those teaching first-year calculus in an ODeL space, this research would benefit from the inclusion of a number of modules in the institution to ascertain profiles and prediction models for each module. Utilising the insights gained from these EDM models, interventions can be designed to address

specific needs and challenges faced by individual students, thereby improving their educational outcomes.

Investigating the generalisability of the model across different institutions and student populations would enhance the robustness of the approach. Replicating this study in traditional, face-to-face settings may also prove fruitful in validating the effectiveness of the framework and decision tree algorithm and assist in identifying potentially at-risk students in different contexts.

The data used to pragmatically demonstrate the KDD process were elementary in nature (consisting of demographic data retrieved during registration and data from the first assignment). Future iterations would benefit from the inclusion of non-cognitive variables, which have been shown to be associated with student success (such as self-efficacy, psychological grit, and motivation (Tladi 2017; K. M. Xu et al. 2020; Müller, Prinsloo, and Du Plessis 2007).

The literature on EDM is replete with studies that incorporate Learning Management System (LMS) data to predict student outcomes in distance education settings (Dietz-Uhler and Hurn 2013; Falakmasir and Habibi 2010; Smith, Lange, and Huston 2012; Macfadyen and Dawson 2012; Minaei-Bidgoli et al. 2003). Considering this, together with the global movement towards online education, future iterations of similar EDM studies would benefit from the inclusion of LMS data.

Conclusion

It is important to note that there may still be a learning curve taking place and the need for some level of technical proficiency. Familiarity with the software interface, understanding the concepts and assumptions underlying the algorithms, and interpreting the results effectively, may require training and practice. Also, collaboration with data science specialists or access to resources that provide guidance and support in using the KDD framework can further enhance the capabilities of non-specialists in utilising data mining techniques for educational purposes in ODeL.

References

Aggarwal, Charu C. 2015. *Data Mining: The Textbook*. Springer International Publishing. https://doi.org/10.1007/978-3-319-14142-8

Alachiotis, N. S., S. Kotsiantis, E. Sakkopoulos, and V.S. Verykios. 2022. “Supervised Machine Learning Models for Student Performance Prediction.” *Intelligent Decision Technologies* 16 (1): 93–106. https://doi.org/10.3233/IDT-210251

Attewell, Paul, and David Monaghan. 2019. *Data Mining for the Social Sciences*. *Data Mining for the Social Sciences*. https://doi.org/10.1525/9780520960596

Aydoğdu, Şeyhmus. 2020. “Predicting Student Final Performance Using Artificial Neural Networks in Online Learning Environments.” *Education and Information Technologies* 25 (3): 1913–27. https://doi.org/10.1007/s10639-019-10053-x

Azevedo, Ana, and Manuel Filipe Santos. 2008. “KDD, SEMMA AND CRISP-DM: A Parallel Overview.” *IADIS European Conference Data Mining*, 182–185.

Bakhshinategh, Behdad, Osmar R. Zaiane, Samira ElAtia, and Donald Ipperciel. 2018. “Educational Data Mining Applications and Tasks: A Survey of the Last 10 Years.” *Education and Information Technologies* 23 (1): 537–53. https://doi.org/10.1007/s10639-017-9616-z

Barnes, Glen. 2014. “‘At-Risk’ Modules per College for Face-to-Face Tuition in 2015.” *Department of Institutional Statistics and Analysis*. Pretoria, University of South Africa (UNISA).

Bell, Gordon, Tony Hey, and Alex Szalay. 2009. “Computer Science: Beyond the Data Deluge.” *Science*. https://doi.org/10.1126/science.1170411

Bendjebar, S., Y. Lafifi, and H. Seridi-Bouchelaghem. 2022. “An Improvement of a Data Mining Technique for Early Detection of At-Risk Learners in Distance Learning Environments.” *International Journal of Knowledge and Learning* 15 (2): 185–202. https://doi.org/10.1504/IJKL.2022.121958

Chen, Eric Evan, and Sean P. Wojcik. 2016. “A Practical Guide to Big Data Research in Psychology.” *Psychological Methods* 21 (4): 458–74. https://doi.org/10.1037/met0000111

Cherkassky, Vladimir, and Filip Mulier. 2006. *Learning from Data: Concepts, Theory, and Methods: Second Edition*. *Learning from Data: Concepts, Theory, and Methods: Second Edition*. John Wiley and Sons. https://doi.org/10.1002/9780470140529

Dietz-Uhler, B., and J. Hurn. 2013. “Using Learning Analytics to Predict (and Improve) Student Success: A Faculty Perspective.” *Journal of Interactive Online Learning* 12 (1): 17–26.

Falakmasir, Mohammad Hassan, and Jafar Habibi. 2010. “Using Educational Data Mining Methods to Study the Impact of Virtual Classroom in e-Learning.” *Proceedings of EDM*, 241–48.

Field, A. 2018. *Discovering Statistics Using IBM SPSS Statistics*. 5th ed. SAGE Publishers.

Fynn, Angelo, and Jaroslaw Adamiak. 2018. “A Comparison of the Utility of Data Mining Algorithms in an Open Distance Learning Context.” *South African Journal of Higher Education* 32 (4): 1–14. https://doi.org/10.20853/32-4-2473

García, S., J. Luengo, and F. Herrera. 2015. *Data Preprocessing in Data Mining*. Springer International Publishing. https://doi.org/https://doi.org/10.1007/978-3-319-10247-4

Jain, Nikita, and Vishal Srivastava. 2013. “Data Mining Techniques: A Survey Paper.” *International Journal of Research in Engineering and Technology* 2 (11): 116–19. https://doi.org/10.15623/ijret.2013.0211019

Janiesch, Christian, Patrick Zschech, and Kai Heinrich. 2021. “Machine Learning and Deep Learning.” *Electronic Markets* 31 (3): 685–95. https://doi.org/10.1007/S12525-021-00475-2

Kurgan, Lukasz A., and Petr Musilek. 2006. “A Survey of Knowledge Discovery and Data Mining Process Models.” *Knowledge Engineering Review* 21 (1): 1–24. https://doi.org/10.1017/S0269888906000737

Lagman, A., and S. Ambat. 2015. “Predictive Analytics of Student Graduation Using Logistic Regression and Decision Tree Algorithm.” In *Proceedings of Second International Conference on Digital Information Processing, Data Mining, and Wireless Communications (DIPDMWC)*, 41–48. Dubai, UAE.

Liu, Ling, and M. Tamer Özsu. 2018. “Encyclopedia of Database Systems.” Edited by Ling Liu and M. Tamer Özsu. *Encyclopedia of Database Systems*. New York, NY: Springer. https://doi.org/10.1007/978-1-4899-7993-3

Macfadyen, Leah P., and Shane Dawson. 2012. “Numbers Are Not Enough. Why e-Learning Analytics Failed to Inform an Institutional Strategic Plan.” *Educational Technology and Society* 15 (3): 149–63. https://doi.org/10.1.1.441.9712

Maimon, O., and L. Rokach. 2011. “Data Mining and Knowledge Discovery Handbook.” *Choice Reviews Online* 48 (10). https://doi.org/10.5860/choice.48-5729

Mason, H., and C. Wiggins. 2010. “A Taxonomy of Data Science.” *Dataists*. 2010. http://www.dataists.com/2010/09/a-taxonomy-of-data-science/

Mccue, Colleen. 2015. *Data Mining and Predictive Analysis: Intelligence Gathering and Crime Analysis*. Second. Elsevier. https://doi.org/10.1016/B978-0-12-800229-2.00004-3

Minaei-Bidgoli, B., D. A. Kashy, G. Kortemeyer, and W. F. Punch. 2003. “Predicting Student Performance: An Application of Data Mining Methods with an Educational Web-based System.” *33rd Annual Frontiers in Education, 2003. FIE 2003.* 1: T2A_13-T2A_18. https://doi.org/10.1109/FIE.2003.1263284

Müller, Heléne, Paul Prinsloo, and Annekie Du Plessis. 2007. “Validating the Profile of a Successful First Year Accounting Student.” *Meditari Accountancy Research*. https://doi.org/10.1108/10222529200700002

Namoun, Abdallah, and Abdullah Alshanqiti. 2021. “Predicting Student Performance Using Data Mining and Learning Analytics Techniques: A Systematic Literature Review.” *Applied Sciences* 11 (1): 1–28. https://doi.org/10.3390/app11010237

Pal, Jiban K. 2011. “Usefulness and Applications of Data Mining in Extracting Information from Different Perspectives.” *Annals of Library and Information Studies* 58.

Ripley, Brian D. 2014. *Pattern Recognition and Neural Networks*. *Pattern Recognition and Neural Networks*. https://doi.org/10.1017/CBO9780511812651

Romero, Cristóbal, and Sebastian Ventura. 2010. “Educational Data Mining: A Review of the State of the Art.” *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 40 (6): 601–618. https://doi.org/10.1109/TSMCC.2010.2053532

Romero, Cristóbal, Sebastian Ventura, Mykola Pechenizkiy, and Ryan Baker. 2011. *Handbook of Educational Data Mining*. Edited by Vipin Kumar. CRC Press. https://doi.org/10.1201/b10274

Salinas, José Gerardo Moreno, and Christopher Stephens. 2015. “Applying Data Mining Techniques to Identify Success Factors in Students Enrolled in Distance Learning: A Case Study.” In *Advances in Artificial Intelligence and Its Applications* 1: 208–19. Morelos, Mexico. https://doi.org/10.1007/978-3-642-45114-0

Shafique, Umair, and Haseeb Qaiser. 2014. “A Comparative Study of Data Mining Process Models (KDD, CRISP-DM and SEMMA).” *International Journal of Innovation and Scientific Research* 12 (1): 217–22.

Slater, Stefan, Srećko Joksimović, Vitomir Kovanovic, Ryan S. Baker, and Dragan Gasevic. 2017. “Tools for Educational Data Mining.” *Journal of Educational and Behavioral Statistics* 42 (1): 85–106. https://doi.org/10.3102/1076998616666808

Smith, Vernon C., Adam Lange, and Daniel R. Huston. 2012. “Predictive Modeling to Forecast Student Outcomes and Drive Effective Interventions.” *Journal of Asynchronous Learning Networks* 16 (3): 51–61. https://doi.org/10.24059/olj.v16i3.275

Song, Yan-Yan, and Ying Lu. 2015. “Decision Tree Methods: Applications for Classification and Prediction.” *Shanghai Archives of Psychiatry* 27 (2): 130–35. https://doi.org/10.11919/j.issn.1002-0829.215044

Tladi, L. S. 2017. “Perceived Ability and Success: Which Self-efficacy Measures Matter? A Distance Learning Perspective.” *Open Learning* 32 (3): 243–61. https://doi.org/10.1080/02680513.2017.1356711

Uni-telligence Data Hub. 2022. *Department of Institutional Intelligence*. Pretoria, University of South Africa (UNISA).

Witten, Ian H., and Eibe Frank. 2005. *Data Mining: Practical Machine Learning Tools and Techniques*. *Complementary Literature None*. Morgan Kaufmann Publishers.

Xu, Kate M., Celeste Meijs, Hieronymus J.M. Gijselaers, Joyce Neroni, and Renate H.M. de Groot. 2020. "Measuring Perseverance and Passion in Distance Education Students: Psychometric Properties of the Grit Questionnaire and Associations With Academic Performance." *Frontiers in Psychology* 11: 1–12. https://doi.org/10.3389/fpsyg.2020.563585

Xu, Y., M. Zhang, and Z. Gao. 2020. "The Construction of Distance Education Personalized Learning Platform Based on Educational Data Mining." In *Applications and Techniques in Cyber Intelligence*, 1076–85. Springer International Publishing. https://doi.org/10.1007/978-3-030-25128-4_134

Yasmin, D. 2013. "Application of the Classification Tree Model in Predicting Learner Dropout Behaviour in Open and Distance Learning." *Distance Education* 34 (2): 218–31. https://doi.org/10.1080/01587919.2013.793642

Zhao, Yanchang, and Yonghua Cen. 2013. *Data Mining Applications with R*. *Data Mining Applications with R*. https://doi.org/10.1016/C2012-0-00333-X